Math Behind

Multiple Linear Regression

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$$

$\downarrow$ ← Target col.

cgpa | iq | gender | lpa

(-> assume there are 100 rows in Data

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_{100} \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \beta_3 x_{13} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \beta_3 x_{23} \\ \vdots & \vdots & \vdots & \vdots \\ \beta_0 & \beta_1 x_{1001} & \beta_2 x_{1002} & \beta_3 x_{1003} \end{bmatrix}$$

$$\hat{y}_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \beta_3 x_{13}$$

* if we had n rows and m cols

$$\hat{Y} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \cdots & \beta_m x_{1m} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \cdots & \beta_m x_{2m} \\ \vdots & \vdots & & & \vdots \\ \beta_0 & \beta_1 x_{n1} & \beta_2 x_{n2} & \cdots & \beta_m x_{nm} \end{bmatrix}$$

* This matrix can be rewrite as

$$X \rightarrow \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1m} \\ 1 & x_{21} & x_{22} & \cdots & x_{2m} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nm} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{bmatrix} \leftarrow \beta$$

A multiplied by B = AB

$$\boxed{\hat{Y} = X\beta} \quad \ldots\ldots \quad (i)$$

-> we can get $\hat{Y}$ by multiplyin X and β
-> now we will se how to find β

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$ <- Target Column Matrix

$$e = Y - \hat{Y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}$$

$$e = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

* in single LR

$$E = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

↑ ~~Loss~~ error func

<- This same formula can be written As :

$$E = e^T e$$

-> we will prove it

$$e^T = \begin{bmatrix} (y_1-\hat{y}_1) & (y_2-\hat{y}_2) & \cdots & (y_n-\hat{y}_n) \end{bmatrix}_{1\times n}$$

$$e = \begin{bmatrix} (y_1-\hat{y}_1) \\ (y_2-\hat{y}_2) \\ \vdots \\ (y_n-\hat{y}_n) \end{bmatrix}_{n\times 1}$$

=> when we multiple This Two matrix we will get single number

$$(1\times n)(n\times 1) = 1\times 1$$

$$e^Te = (y_1-\hat{y}_1)^2 + (y_2-\hat{y}_2) + (y_3-\hat{y}_3) + \dots + (y_n-\hat{y}_n)$$

$$e^Te = \sum_{i=1}^{n} (y_i-\hat{y}_i)^2$$

$$\boxed{e = (Y-\hat{Y})}$$

$$E = e^Te = (Y-\hat{Y})^T(Y-\hat{Y})$$

$$* \quad (A-B)^T = A^T - B^T$$

$$E = (Y^T-\hat{Y}^T)(Y-\hat{Y})$$

=> we will use $\hat{Y} = X\beta$ in this Term

$$E = \left[Y^T - (X\beta)^T\right] \cdot (Y - X\beta)$$

$$E = Y^T Y - Y^T X\beta - (X\beta)^T Y + (X\beta)^T X\beta$$

* we have To Prove This Two as equal

Means: $Y^T X\beta = (X\beta)^T Y$

* Lets Prove it

$$Y^T X\beta = (X\beta)^T Y$$

- Assume $Y = A$, $X\beta = B$

$$A^T B = B^T A \quad \ldots (i)$$

- Now what will be ...

$(A^T B)^T = ?$ $\qquad (AB)^T = B^T A^T$

$(A^T B)^T = B^T A \ldots (ii)$ ⟵ By this Logic

* $A^T B = B^T A \quad \ldots (i)$

$(A^T B)^T = B^T A \quad \ldots (ii)$

-> in both Term RHS is crucial
So we have To Prove..

$$(A^TB)^T = A^TB$$

assume $A^TB = C$

$\therefore C = C^T$

$C = A^TB$

$A = Y^T$
$B = X\beta$

* we have to Prove That

$$(Y^T X\beta)^T = (Y^T X\beta)$$

$$Y = \begin{bmatrix} - \\ - \\ - \end{bmatrix}_{(n\times 1)} \text{ (Target Col)}$$

Y^T shape $= (1 \times n)$

X = feature + 1 col. containing 1

X shape $= (n \times (m+1))$

β = Coefficient
β's shape $= (m+1) \times 1$

shape of

$Y^T = (1 \times n)$

$X = (n \times (m+1))$

$\beta = (m+1) \times 1$

so shape of

$(Y^T X \beta)^T$ will be

$(1 \times n)\,(n \times (m+1))\,[(m+1) \times 1]$

$[1 \times (m+1)]\,[(m+1) \times 1]$

1×1

Transpose of matrix having 1×1 shape will be the matrix itself

for ex :- $[7]^T = [7]$

we proved :- $Y^T X \beta = (X\beta)^T Y$

* we came back To E where

$$E = Y^T Y - Y^T X\beta - (X\beta)^T Y + (X\beta)^T X\beta$$

$$E = Y^T Y - 2(Y^T X\beta) + \beta^T X^T X\beta$$

-> This our final Error (Loss) function.

-> lets minimize loss function

$$\frac{dE}{d\beta} = \frac{d}{d\beta}\left[Y^T Y - 2Y^T X\beta + \beta^T X^T X\beta \right] = 0$$

$$= 0 - 2Y^T X + \frac{d}{d\beta}\underbrace{\left[\beta^T X^T X\beta\right]}_{\text{matrix differentiation}} = 0$$

$$= -2Y^T X + 2X^T X\beta^T = 0$$

$$= \cancel{2}X^T X\beta^T = \cancel{2}Y^T X$$

RHS

$$\beta^T = Y^T X (X^T X)^{-1}$$

$$(\beta^T)^T = \left[Y^T X (X^T X)^{-1} \right]^T$$

$$\beta = \left[(X^T X)^{-1}\right]^T (Y^T X)^T.$$

$$\beta = \left[(X^T X)^{-1}\right]^T (X^T Y)$$

Transpose of square matrix

$$\beta = (X^T X)^{-1} X^T Y$$

$$\boxed{\beta = (X^T X)^{-1} X^T Y}$$

* $X \rightarrow$ X_train + $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

* $Y \rightarrow$ Y_train

* Shape :-

$$\beta = (X^T X)^{-1} X^T Y$$

$$(m+1) \times 1 = \left[(m+1) \times (m+1)\right]\left[(m+1) \times n\right]\left[n \times 1\right]$$

$$\left[(m+1) \times n\right]\left[n \times 1\right]$$

$$(m+1) \times 1 = (m+1) \times 1$$